Joint Adversarial Network With Semantic and Topology Fusion for Cross-Scene Hyperspectral Image Classification

Ronghua Shang, *Senior Member, IEEE*, Yuhao Xie, Weitong Zhang, *Member, IEEE*, Jie Feng, *Member, IEEE*, and Songhua Xu

***Abstract*— Hyperspectral image cross-scene classification (HSICC) poses a significant challenge due to distribution variations between source and target domains. Existing unsupervised domain adaptation methods primarily focus on local knowledge transfer, often neglecting the critical semantic information and sample topological structure inherent in hyperspectral images (HSIs). To address these limitations, this article introduces an end-to-end joint adversarial network with semantic and topology fusion (JAN-STF). This network liberates from the constraints of local perception by integrating semantic and topological information into both domain- and class-level adversarial learning processes. First, the network constructs a semantic-guided cross-domain graph structure to obtain cross-domain features. Subsequently, domain-level adversarial learning is conducted using these features to achieve domain-invariant representation with robust transferability. Moreover, to bolster stability in the ensuing class-level adversarial procedure, the network dynamically computes cross-domain category center distance loss utilizing an intra-domain topological semantic attention mechanism, thereby mapping features to proximate spaces. Finally, class-level adversarial learning is performed by leveraging the prediction discrepancy between the local classifier and the topological classifier, thus enhancing the discriminative performance of the domain-invariant representation. Extensive experiments on three broadly utilized HSICC datasets demonstrate JAN-STF's superiority in accuracy and Kappa coefficient (KC) metrics over nine leading algorithms.**

***Index Terms*— Adversarial learning, domain adaptation, hyperspectral image (HSI), semantic attention, topological structure.**

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I. Introduction

HYPERSPECTRAL images (HSI) are widely utilized in various fields such as natural disaster prediction and environmental monitoring due to their rich spectral information [1]. The applications of HSIs encompass diverse tasks, including denoising [2], target detection [3], and pixel-level classification [4]. Pixel-level land cover classification in hyperspectral remote sensing [5] is essential for extracting valuable surface information and providing insights for decision-making in various fields like environmental protection and urban development planning.

In recent years, deep learning methods have gained widespread popularity in HSI classification (HSIC). This includes approaches using convolutional neural networks (CNN) [6], methods leveraging deep encoding mechanisms [7], and techniques incorporating attention mechanisms [8]. However, these HSIC techniques typically require a large number of annotated samples for satisfactory classification results and are constrained to specific scenarios. Acquiring ample labeled data is costly and time-consuming in real-world settings, resulting in a shortage of available labeled samples, whereas unlabeled samples are more prevalent [9].

Transfer learning in HSI cross-scene classification (HSICC) offers an effective solution to the mentioned challenges. It leverages labeled samples from similar scenes (domains) during training, transferring knowledge to unlabeled target scenes (domains) [10]. However, variations in feature distributions across scenes persist due to various factors such as data acquisition devices and spectral responses. Domain adaptation (DA), a submethod of transfer learning, effectively addresses domain bias challenges, especially with identical categories across scenes, enhancing classification performance and generalization in the target scene. Unsupervised domain adaptation (UDA) has gained increased attention for not needing labeled samples from the target domain in DA tasks [11].

Compared to traditional UDA methods [12], deep domain adaptation methods have gained prominence with the emergence of deep neural networks. Some scholars attempt to approach domain adaptation from the perspective of deep network structures. Li et al. [13] proposed adaptive batch normalization (AdaBN), which applies batch normalization sequentially on the source and target domains. Moreover, the concepts of self-supervised, few-shot learning

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and meta-learning are currently widely utilized in HSICC. Wang et al. [14] proposed a DA framework, which integrates contrastive self-supervised learning and consistent self-training. Zhao et al. [15] proposed a locally linear unbiased randomization network (LLURnet) for HSICC, which leverages self-supervised learning and adversarial training to enhance classification on the target domain. Wang et al. [16] developed a cross-domain few-shot learning framework based on class attention mechanism and asymmetric adversarial training to learn domain-invariant features. Cheng et al. [17] proposed a meta-learning-based method that integrates meta-learning with causal modeling. Gao et al. [18] introduced an unsupervised meta-learning-based multiview constraint method to enhance the classification performance of hyperspectral data. Nevertheless, these approaches often necessitate the tailoring of specific network architectures or learning strategies for diverse tasks, thereby presenting limitations in terms of overall generality.

Hence, another prevalent research direction focuses on aligning data distributions in domain adaptation to minimize differences across domains using explicit or implicit metrics. Maximum mean discrepancy (MMD) [19] as an explicit metric has been widely applied in various deep domain adaptation studies. Long et al. [20] introduced a deep adaptation network (DAN) using a multikernel MMD (MK-MMD) metric to learn statistically transferable features. In HSICC, Zhang et al. [21] proposed a topological structure and semantic information transfer network (TSTnet), which aligned the distribution of local and intra-domain topological information using two explicit metrics, MMD and graph optimal transport. Qi et al. [22] introduced a similarity-value progressive constraint based on explicit cosine distance to explore the intrinsic attributes within the domain and cross-domain common features. However, explicit metrics require manual selection and struggle with high-dimensional nonlinear features.

Diverging from explicit function, implicit metrics from adversarial domain adaptation (ADA) exhibit greater flexibility and adaptability. Typically, the current implementations of ADA exist in three manners, all of which have been widely applied in HSICC tasks. The first approach is the domain-level adversarial learning proposed by [23]. This adversarial strategy has also been commonly integrated with style transfer methods [24] in recent years. Li et al. [25] introduced a DualGAN-based style transfer method for aerial image classification. Luo et al. [26] presented an adversarial style discrepancy minimization method to achieve fine-grained domain adaptation through adversarial training. For HSICC, Meng et al. [27] introduced a cycle-assisted classification generative adversarial network, achieving bidirectional mapping and alignment across domains. The second form of ADA is class-level adversarial learning [28]. This adversarial strategy also synergizes with self-training methods [29] that have shown remarkable performance in UDA recently. This stems from that self-training method may overfit to certain categories due to uncertainty from unsupervised-generated pseudo-labels, especially in noisy or imbalanced scenarios [30]. Adversarial learning can effectively suppress these negative effects brought by erroneous pseudo-labels at the feature level [31]. Capliez et al. [32] introduced a framework that leverages adversarial learning and self-training for temporal unsupervised domain adaptation. Wang et al. [33] proposed a bi-classifier adversarial network (BCAN) that incorporates self-training strategies, effectively enhancing feature discriminative performance. The third paradigm of ADA is the joint adversarial learning [34], which combines the first and second forms of ADA. Tang et al. [35] proposed an unsupervised joint ADA (UJADA) method that employs CDD to achieve joint adversarial learning in HSICC.

Although these methods have made specific improvements for HSICC tasks, most of them rely solely on locally extracted features from CNN, overlooking the rich topological and semantic information in HSI that contributes to knowledge transfer. The topological structure can effectively characterize the abundant spatial information of HSI, while semantic information provides meaningful representations of the intricate spectral information. Moreover, current ADA methods for HSICC task lack research and advancements in maintaining adversarial stability. As a classic domain adaptation algorithm, ADA excels at capturing high-order features and structures within data distributions, enabling adaptation to nonlinear domain variations. However, it faces challenges of training instability, particularly evident when dealing with highly complex spectral features in HSI.

To address the aforementioned limitations, this article proposes an end-to-end joint adversarial network with semantic and topology fusion (JAN-STF). JAN-STF utilizes intra-domain and cross-domain topological structures to fully leverage the spatial information in HSI, while employing semantic attention mechanisms to transform spectral information into semantic labels with supervisory information. These semantic and topological information are integrated into adversarial learning to overcome limitations in local knowledge transfer. The integration is conducted at both domain- and class-levels, enhancing the performance of domain adaptation and the stability of learning. Specifically, JAN-STF first employs a semantic attention mechanism, combining local features extracted by CNN to construct a cross-domain topological structure guided by class semantics. This graph structure utilizes graph convolutional networks (GCNs) to aggregate distribution information from both the source and target domains. It facilitates seamless information exchange between scenes, generating cross-domain features with greater transferable potential for domain-level adversarial learning. Conducting domain-level adversarial learning based on these features can acquire domain-invariant representations with strong transferability. Furthermore, a semantic attention mechanism, incorporating intra-domain topology information, generates semantic pseudo-labels. Coupled with source labels, it dynamically calculates the distance loss of category centers across domains. This topology-aware category alignment strategy seeks to project features of the same categories in two domains to nearby spaces, maintaining stability in subsequent class-level adversarial process. Finally, two similar

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yet distinct classifiers are constructed using local features from CNN and intra-domain topological structure information from GCN. Leveraging the significant dissimilarity between the two classifiers, class-level adversarial learning is performed based on the disparity in decision boundaries. This effectively integrates topological information into local features, further enhancing the discriminative performance of domain-invariant representations for high-performance classification in target scenario.

The principal contributions of this article are delineated as follows.

1) A cross-domain topological structure guided by category semantics for aggregating cross-domain data distribution information is introduced. It breaks down barriers of information exchange across scenes, producing cross-domain features with greater transferable potential for domain-level adversarial learning.
2) A dynamic category alignment mechanism based on intra-domain topological semantic attention is proposed. This mechanism effectively reduces the feature distance between cross-domain same-class instances to a relatively stable range, maintaining the stability of subsequent class-level adversarial learning.
3) A dual-classifier adversarial paradigm based on local features and intra-domain topological information is developed. It effectively integrates topological information with local features, capturing more distant target samples and enhancing the discriminative performance of domain-invariant representation.

The following sections of this article are organized as detailed below. Section II provides an overview of works closely related to this study. Section III furnishes an elaborate exposition of the proposed algorithm. In Section IV, the experiment results on three benchmark datasets are presented. The concluding remarks are encapsulated in Section V.

II. Related Work

This section primarily introduces methods closely related to this work, namely ADA and graph convolution in HSI.

A. Adversarial Domain Adaptation

In the realm of adversarial domain adaptation (ADA), contemporary methods can be broadly categorized into three types based on the adversarial forms: domain-level, class-level, and joint adversarial learning. Ganin et al. [23] first proposed the domain-level adversarial learning method, namely domain adversarial neural network (DANN). Within the DANN framework, the feature extractor aimed to extract domain-invariant features, while the domain discriminator endeavored to distinguish the features from the source and target domains. Building upon this idea, Long et al. [36] introduced a conditional ADA (CDAN) framework which employed two novel conditional strategies to align diverse domains in multimodal distributions. However, domain-level adversarial learning may result in the degradation of feature discriminability, leading to negative alignment. Therefore, Saito et al. [28] initially proposed the second form of ADA, namely class-level adversarial learning, to enhance the discriminative quality of domain-invariant representations. The method utilizes maximum classifier discrepancy (MCD) for an adversarial game between two classifiers and feature extractors. Here, the feature extractor minimized predictive discrepancy between the classifiers, while the classifiers strived to maximize it. Li et al. [37] advanced a bi-classifier determinacy maximization (BCDM) method to encourage the generation of distinctive feature representations, thereby maintaining diversity within an adversarial framework. Nevertheless, class-level adversarial learning, while focusing on class alignment, faces challenges in addressing significant domain discrepancies. In contrast, domain-level adversarial learning can effectively reduce overall domain discrepancies. Motivated by this, Li et al. [34] suggested the third adversarial form, joint ADA (JADA) which harnessed the complementary nature of domain-level and class-level adversarial learning, resulting in enhanced domain adaptation. The joint adversarial network serves as the foundational adversarial framework for this work.

B. Graph Convolution in HSI

1) Graph Construction: Currently, prevalent graph convolution methods in HSI can be divided into two categories based on graph construction: pixel-level and superpixel-level methods. Pixel-level methods treat pixels as graph nodes, constructing precise graph networks from original spectral features and spatial relationships. For instance, Hong et al. [38] proposed miniGCN, but this approach incurred high computational costs due to the large number of nodes. In contrast, superpixel-level methods partition the image into superpixel blocks using algorithms like SLIC [39] and perform graph convolution on these blocks as nodes. Representative works include CNN-enhanced GCN by Liu et al. [40] and MGLN by Wan et al. [41], offering lower computational costs but potentially aggregating incorrect class information. To address these challenges, Zhang et al. [21] proposed a dynamic subgraph construction method within the training process, updating topological subgraphs using locally extracted CNN features in each batch, achieving more accurate feature embedding. This dynamic graph construction method is employed in the topological graph network construction in this article.

2) Graph Sampling Aggregation Network: Graph sampling aggregation network (GraphSAGE) [42] is a node representation learning algorithm that generates low-dimensional vectors for each node by aggregating features from neighboring nodes. It iteratively updates node representations using the following formula:

$$h_v^{(k+1)} = \mathcal{A}\left(\left\{h_u^{(k)}, \forall u \in S(v)\right\}\right) \quad \forall v \in V,\ k = 0, 1, \ldots \quad (1)$$

where $h_u^{(k)}$ represents the representation vector of node v after k iterations. $S(v)$ is the subset sampled from the neighbor nodes of node v, V denotes the set of nodes in the graph, and $\mathcal{A}(\cdot)$ is the aggregation function used to aggregate the representation vectors of sampled neighbor nodes. GraphSAGE is used for information aggregation across all topological structures in this article.

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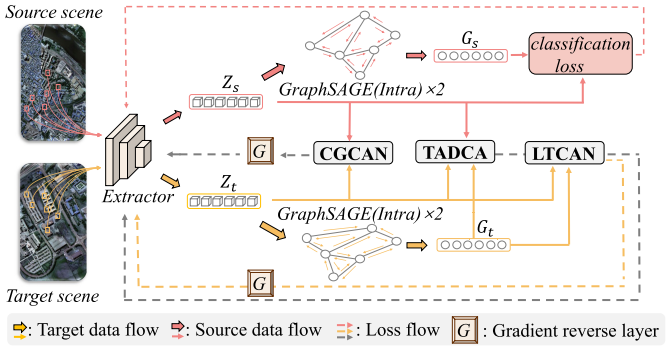


Fig. 1. Framework of JAN-STF.

III. Proposed Method

To address the limitations of current domain adaptation methods in the HSICC task, which are constrained by the limited modeling capacity of CNN, this article proposes JAN-STF. The framework of JAN-STF is illustrated in Fig. 1.

As shown in Fig. 1, JAN-STF comprises two main parts: feature generation at the top and bottom, and domain adaptation in the middle. Feature generation includes extracting local and intra-domain topological features. The domain adaptation consists of three components: cross-graph convolution adversarial network (CGCAN), topology-aware dynamic class alignment (TADCA), and local-topology classifier adversarial network (LTCAN). Initially, JAN-STF conducts domain-level adversarial learning via CGCAN to achieve a domain-invariant representation with strong transferability. Subsequently, TADCA reduces the distance between cross-domain class features, reducing the risk of collapse in class-level adversarial procedures. Finally, LTCAN enhances the discriminative ability of the domain-invariant representation through class-level adversarial learning, yielding a highly discriminative and transferable cross-domain feature representation for superior classification performance in the target domain. The following sections will provide detailed explanations of the three components of JAN-STF domain adaptation.

A. Domain-Invariant Representation Learning via CGCAN

Conventional domain-level adversarial networks [23] may lack sufficient transferability in extracted cross-domain features due to their reliance on local features from a single scene. To overcome this limitation and achieve features with stronger transferability, we propose a cross-scene topological structure guided by semantic information, as depicted in Fig. 2.

The cross-scene topological structure, as shown in Fig. 2, integrates nodes representing three land cover class pairs (tree, bare soil, and asphalt) from the source scene with corresponding pseudo-labeled (PL) nodes in the target scene, guided by semantic labels. This structure overcomes local feature limitations within a single scene, effectively amalgamating spectral data from both scenes and, in conjunction with domain-level adversarial learning, discovers domain-invariant features with improved generalization and expressive capabilities across both scenes.

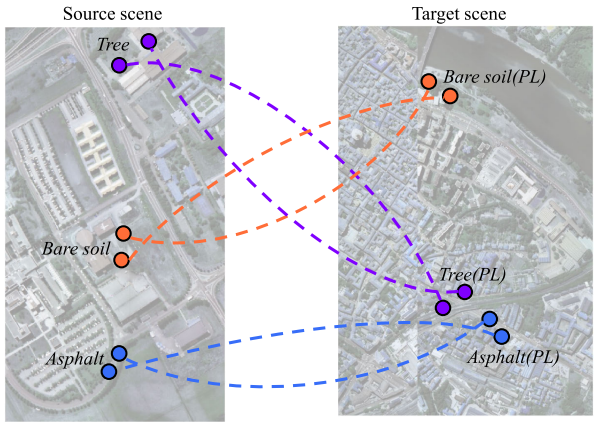


Fig. 2. Semantic-guided cross-scene topological structure diagram.

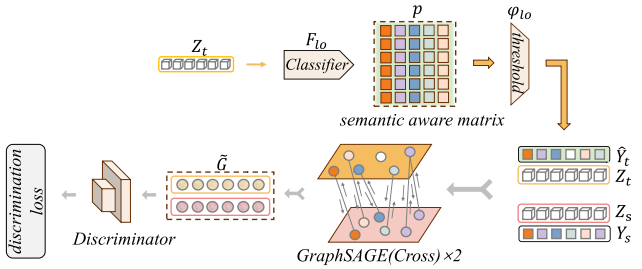


Fig. 3. Structure of the CGCAN.

In JAN-STF, CGCAN is the concrete realization of the aforementioned idea. Specifically, for the HSICC task addressed in this article, the setup is similar to other UDA classification tasks. It involves a sample set $X_s = \{x_i^s\}_{i=1}^{n_s}$ with corresponding labels $Y_s = \{y_i^s\}_{i=1}^{n_s}$ from the source domain and an unlabeled target domain sample set $X_t = \{x_j^t\}_{j=1}^{n_t}$. Here, x_i^s and x_j^t represent pixel blocks in images from the source and target scenes, respectively, and y_i^s is the corresponding class label for x_i^s. The quantities n_s and n_t denote the number of samples in the source and target domains, respectively. Next, after the images from the source and target scenes undergo mapping by the feature extractor, we obtain local features $Z_s = \{z_i^s\}_{i=1}^{n_s}$ and $Z_t = \{z_i^t\}_{i=1}^{n_t}$. CGCAN then performs domain-level adversarial learning on Z_s and Z_t, and its specific structure is illustrated in Fig. 3.

From Fig. 3, it can be seen that initially, the local features with rich but intricate spectral information of the target domain Z_t are input into the corresponding local feature classifier F_{lo}. This process yields a semantic attention matrix, denoted as p, based on prediction scores, with dimensions of $n_t \times c$ where c represents the overall count of categories. The formal depiction of this operation is given by

$$p = \left\{F_{\text{lo}}\left(z_i^t\right)\right\}, \quad i = 1, \ldots, n_t \tag{2}$$

where z_i^t is a local feature in Z_t. Subsequently, the pseudo-label is assigned to the ith target domain sample using the following equation:

$$\hat{y}_i^t = \begin{cases} \underset{j}{\text{argmax}}\, p^i, & p_j^i > \varphi_{\text{lo}} \\ -1, & \text{otherwise} \end{cases} \tag{3}$$

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where p_j^i represents the estimated score of the ith sample being predicted as class j based on z_i^t. When the maximum score in p^i is greater than the predefined local threshold φ_{lo}, the class corresponding to that score is assigned as the semantic pseudo-label for this sample. Conversely, -1 indicates that the sample cannot be assigned a pseudo-label. After applying the thresholding strategy to all target domain samples, the poor-quality incorrect pseudo-labels will be removed, resulting in a high-quality pseudo-label $\hat{Y}_t = \{y_i^t\}_{i=1}^{n_t}$. This strategy effectively prevents the model from negative learning due to erroneous pseudo-labels, especially during the early stages of training. Subsequently, a cross-domain graph network is constructed by combining the local features of the source and target domains and acquired label information, and information aggregation is performed using GCN. This GCN operation can effectively capture the abundant spatial relationships across domains which ultimately yields the cross-domain topological features $\widetilde{G} = \{\tilde{g}_i\}_{i=1}^{n_s+n_t}$. During the cross-domain graph propagation process, the cross-domain topological features are initialized with Z_s and Z_t, and embedded into node representations. Following this, two layers of GraphSAGE [42] are employed for cross-domain information propagation. The convolutional update process for each layer is defined as follows:

$$\tilde{g}_i^l = \sigma\left(\widetilde{W}^{l-1} \cdot \text{Mean}\left(\begin{matrix}\{\tilde{g}_i^{l-1}\} \\ \cup \\ \{\tilde{g}_k^{l-1} \forall k \in \widetilde{\mathcal{N}}(i)\}\end{matrix}\right)\right) \tag{4}$$

where i represents the current node index, and k denotes the index of the node sampled as a neighbor to the ith node through the cross-domain sampling function $\widetilde{\mathcal{N}}(i)$. The $\cup$ represents the union operation and the Mean$(\cdot)$ operation here denotes the averaging of features between the current node and its neighboring nodes. $\widetilde{W}^{l-1}$ represents the parameters of the cross-domain GraphSAGE layer at the lth layer, $\sigma(\cdot)$ represents the nonlinear activation function, and $\tilde{g}_i^l$ represents the output of the lth layer of GraphSAGE. Additionally, the definition of the cross-domain sampling function $\widetilde{\mathcal{N}}(i)$ is as follows:

$$\widetilde{\mathcal{N}}(i) = \text{RandomSelect}\left(\left\{j, \left(\widetilde{A}_{ij} = 1\right)\right\}\right), \quad j = 1, \ldots, n_s + n_t \tag{5}$$

where $\widetilde{A}$ represents the adjacency matrix of the cross-domain topological relation graph, employing semantic category connections. For instance, $\widetilde{A}_{ij} = 1$ indicates that the ith node in the source domain shares the same category as the jth node in the target domain, meaning $y_i^s = \hat{y}_j^t$. Therefore, these two nodes are considered to be connected. Conversely, $\widetilde{A}_{ij} = 0$ implies that there is no connection between the two nodes. Finally, the feature extractor engages in an adversarial game with the domain discriminator, leveraging cross-domain topological features. The discriminator's task is to precisely distinguish between source and target domain features, optimizing itself based on the following equation:

$$\min_{\theta_D} \mathcal{L}_{\text{discrim}}\left(\widetilde{G}\right) \tag{6}$$

where D is the domain discriminator, and θ_D represents the training parameters of the domain discriminator. $\mathcal{L}_{\text{discrim}}$ denotes the discrimination loss in the domain discriminator, which can be computed using the following equation:

$$\mathcal{L}_{\text{discrim}}\left(\widetilde{G}\right) = -\frac{1}{n_s}\sum_{i=1}^{n_s}\log\left(D\left(\tilde{g}_i\right)\right) - \frac{1}{n_t}\sum_{i=n_s+1}^{n_s+n_t}\log\left(1 - D\left(\tilde{g}_i\right)\right) \tag{7}$$

where the log represents the logarithm operation. From (7), it can be seen that the optimization formula in (6) is based on the binary classification loss of the domain discriminator. In contrast, the objective of the feature extractor is to extract features that can confuse the domain discriminator, and thus, it can be optimized using the following equation:

$$\min_{\theta_E} -\mathcal{L}_{\text{discrim}}\left(\widetilde{G}\right) \tag{8}$$

where θ_E represents the training parameters for the feature extractor E. By iteratively engaging in this adversarial game, the overall distribution discrepancy between the two domains can gradually be reduced, leading to the acquisition of domain-invariant representations with strong transferability.

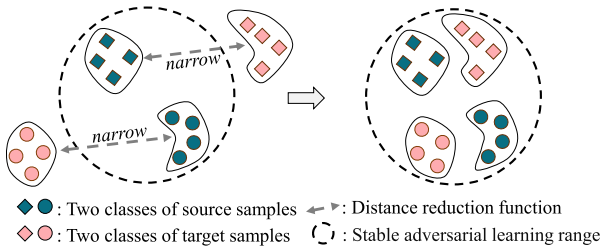


Fig. 4. Diagram illustrating the process of narrowing cross-domain features' distribution distance within the same category.

B. Class Alignment via TADCA

Adversarial domain-level learning may decrease feature discriminability, leading to negative alignment in cross-domain category features. Class-level adversarial training is commonly used to enhance feature discriminability. However, HSIs from different scenes may show significant distribution differences even within the same category due to various acquisition conditions. Direct class-level adversarial training could result in the collapse phenomenon, where target samples deviate too far from the decision boundary supported by the source domain. To address this, we introduce the TADCA mechanism before class-level adversarial learning. TADCA aligns categories by reducing the distance between features of cross-domain samples from the same category, ensuring stability in subsequent adversarial learning. The process is visualized in Fig. 4.

From Fig. 4, it is evident that the features of two initially distantly distributed classes in the source and target scenes are brought closer through a distance reducing function. This ensures stable feature distributions across scenes, essential for effective adversarial learning, and reduces the risk of class adversarial learning breakdown. Furthermore, unlike previous pseudo-labeling and moving-average-based alignment methods [43], TADCA employs a topology-aware mechanism to

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improve pseudo-label credibility. Since topological features aggregate information from multiple class nodes, pseudo-labels generated based on topological features can more effectively avoid erroneous pseudo-labels caused by domain shifts in categories. TADCA also utilizes dynamic category center computation to minimize error accumulation, effectively mitigating the negative impact of incorrect pseudo-labels and achieving more precise category alignment.

The TADCA mechanism performs class alignment based on Z_s, Z_t, and $G_t = \{g_i^t\}_{i=1}^{n_t}$, where G_t represents the intra-domain topological features of the target domain obtained from the aggregation of topological structure information using GCN. Specifically, the topological features are initialized by Z_t and embedded into node representations. Then, two layers of GraphSAGE, which employ an average aggregation function, are utilized for intra-domain spatial information propagation. The convolutional update process for each layer is as follows:

$$g_i^{t(l)} = \sigma\left(W^{l-1} \cdot \text{Mean}\left(\begin{array}{c}\left\{g_i^{t(l-1)}\right\} \\ \cup \\ \left\{g_k^{t(l-1)} \forall k \in \mathcal{N}(i)\right\}\end{array}\right)\right) \tag{9}$$

where i represents the index of the current node, and k represents the index of the node sampled by the intra-domain sampling function $\mathcal{N}(i)$ that is adjacent to the ith node. $g_i^{t(l)}$ represents the output of the lth layer of GraphSAGE for the target domain, and the weights of the lth layer in GraphSAGE are denoted by W^{l-1}. Equations (9) and (4) are structurally similar, but they differ in node definition. Equation (4) encompasses nodes from both the source and target domains, while (9) only includes nodes from the target domain. Furthermore, (4) employs a cross-domain sampling function, while (9) utilizes an intra-domain sampling function, defined as follows:

$$\mathcal{N}(i) = \text{RandomSelect}\left(\left\{j, \left(A_{ij}^t = 1\right)\right\}\right), \quad j = 1, \ldots, n_t \tag{10}$$

where A^t represents the adjacency matrix between nodes and $A_{ij}^t = 1$ indicates a connection between the ith and jth nodes in the target domain. To be noted is that the topological relationships of the intra-domain refer to the work [21], using a fully connected approach for nodes. Therefore, $\mathcal{N}(i)$ represents a set of randomly selected nodes from those adjacent to the ith node. The aforementioned process facilitates the acquisition of intra-domain topological features G_t for the target domain. Note that the construction process of the source domain topological features $G_s = \{g_i^s\}_{i=1}^{n_s}$ is similar to that of G_t. Subsequently, category alignment can be performed using TADCA, as illustrated in Fig. 5.

From Fig. 5, it can be observed that first, the topological feature G_t of the target domain is fed into the corresponding topological feature classifier F_{to}. This yields a semantic attention matrix q based on prediction scores, with dimensions $n_t \times c$. This process can be formulated as

$$q = \left\{F_{\text{to}}\left(g_i^t\right)\right\}, \quad i = 1, \ldots, n_t \tag{11}$$

where g_i^t is an intra-domain topological feature in G_t. Next, the pseudo-label is assigned to the ith target domain sample

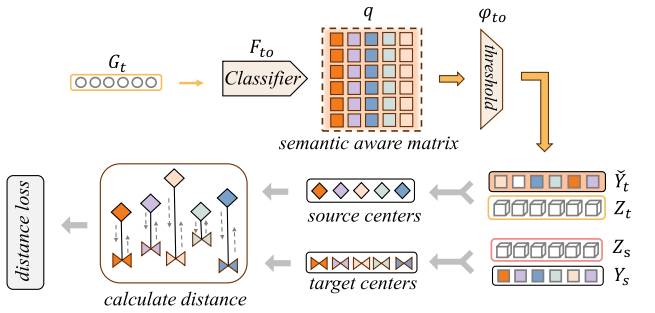


Fig. 5. Structure of the TADCA.

using the following formula:

$$\check{y}_i^t = \begin{cases} \underset{j}{\text{argmax}}\, q^i, & q_j^i > \varphi_{\text{to}} \\ -1, & \text{otherwise} \end{cases} \tag{12}$$

where q_j^i represents the estimated score of the ith sample being predicted as class j based on g_i^t. When the maximum score in q^i is greater than the predefined topological threshold φ_{to}, the corresponding class is assigned as the semantic pseudo-label for this sample; otherwise, -1 is used to indicate that the sample cannot be assigned a pseudo-label. After threshold filtering for all target domain samples, the pseudo-labels $\check{Y}_t = \{\check{y}_i^t\}_{i=1}^{n_t}$ can be obtained. Notably, $\check{Y}_t$ and above $\hat{Y}_t$ are semantic pseudo-labels independently generated by the topological and local classifiers, significantly diversifying pseudo-labels and preserving discriminability between topological and local features. This mitigates the performance decline issue in previous methods [33], which utilized the consistency of same classifier for pseudo-label generation led to biased class distributions during training. Subsequently, inspired by the work [43], this article optimizes the feature extractor by narrowing the category center's distance between the source and target domain categories for achieving cross-domain class distribution alignment. The formulation of this optimization can be expressed as follows:

$$\min_{\theta_E} \mathcal{L}_{\text{dist}}\left(Z_s, Y_s, Z_t, \check{Y}_t\right) \tag{13}$$

where $\mathcal{L}_{\text{dist}}$ is the distance loss. It can be computed by

$$\mathcal{L}_{\text{dist}}\left(Z_s, Y_s, Z_t, \check{Y}_t\right) = \sum_{j=1}^{c} \Phi\left(C_j^s, C_j^t\right) \tag{14}$$

where $\Phi(\cdot, \cdot)$ is the distance function, and in this article, the squared Euclidean distance is employed, defined as $\Phi(x, x') = \|x - x'\|^2$. C_j^s and C_j^t represent the category centers for class j in the source and target domains, respectively, and are computed using the following formula:

$$C_j^s = \text{Mean}\left(\left\{z_i^s, y_i^s = j\right\}\right), \quad i = 1, \ldots, n_s \tag{15}$$

$$C_j^t = \text{Mean}\left(\left\{z_i^t, \check{y}_i^t = j\right\}\right), \quad i = 1, \ldots, n_t \tag{16}$$

where the Mean$(\cdot)$ refers to the averaging of all local features belonging to the same category. This calculation of category centers is updated batch-wise, effectively reducing the cumulative negative impact of incorrect pseudo-labels. Overall, after the optimization as shown in (13), the feature

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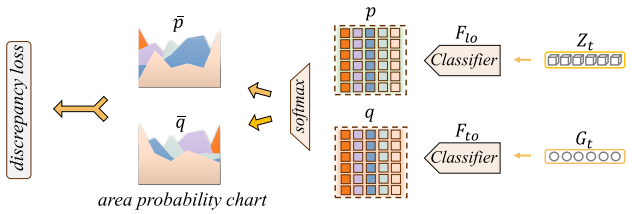


Fig. 6. Structure of the LTCAN.

extractor gradually maps features of cross-domain categories into nearby spaces.

C. Discriminative Learning via LTCAN

Following domain-level adversarial learning that yields a transferable domain-invariant representation, the subsequent step involves conducting class-level adversarial learning via LTCAN to augment the discriminative capability of the domain-invariant representation. Unlike conventional approaches that rely solely on dual classifiers with different initializations [28], LTCAN utilizes the disparity between local and topological classifiers which are based on the local and topological features for class-level adversarial training. Leveraging the distinctiveness between local and topological features enhances the ability to identify target domain samples that are distant from the support of the source domain, capturing richer information from these samples. Throughout the adversarial process, the feature extractor aligns local and topological features effectively, reducing the discrepancy in classifier decisions and ensuring the incorporation of topological structure information into classification decisions.

LTCAN conducts class-level adversarial learning based on the target domain local features Z_t and topological features G_t. The specific structure is illustrated in Fig. 6.

In Fig. 6, the matrices p and q from the local and topological classifiers, respectively, are subjected to softmax along the category dimension, yielding $\bar{p}$ and $\bar{q}$. This step prevents collapse due to diverse feature-based predicted probabilities. Subsequently, the feature extractor is optimized to reduce classifier disparity with the equation

$$\min_{\theta_E} \mathcal{L}_{\text{discrep}}\left(\bar{p}, \bar{q}\right) \tag{17}$$

where $\mathcal{L}_{\text{discrep}}$ is the discrepancy loss between the local and topological classifiers. It can be computed as follows:

$$\mathcal{L}_{\text{discrep}}\left(\bar{p}, \bar{q}\right) = \frac{1}{n_t \times c} \sum_{i=1}^{n_t} \sum_{j=1}^{c} \phi\left(\bar{p}_j^i, \bar{q}_j^i\right) \tag{18}$$

where $\phi(\cdot, \cdot)$ is the distance metric function, and in this article, the L1 distance function is employed, i.e., $\phi(x, x') = |x - x'|$. $\bar{p}_j^i$ and $\bar{q}_j^i$ represent the prediction probability for class j. On the contrary, the classifiers need to amplify the differences in predicted results between two classifiers. Therefore, they are optimized using the following equation:

$$\min_{\theta_{F_{\text{lo}}}, \theta_{F_{\text{to}}}} -\mathcal{L}_{\text{discrep}}\left(\bar{p}, \bar{q}\right) \tag{19}$$

where $\theta_{F_{\text{lo}}}$ and $\theta_{F_{\text{to}}}$ represent the training parameters of the local classifier and the topological classifier, respectively. By iteratively repeating this adversarial game, the discriminative performance of domain-invariant representations can be effectively enhanced, thereby improving the classification performance in the target domain.

D. Training Optimization and Inference

In addition to the mentioned optimization, JAN-STF further optimizes the feature extractor and classifiers using local features Z_s, topological features G_s, and corresponding source domain labels Y_s to stabilize the overall classification performance of the model. The optimization process is as follows:

$$\min_{\theta_E, \theta_{F_{\text{lo}}}, \theta_{F_{\text{to}}}} \mathcal{L}_{\text{cls}}(Z_s, G_s, Y_s) \tag{20}$$

where $\mathcal{L}_{\text{cls}}$ signifies the classification loss of the source domain samples. It is calculated through the following equation:

$$\mathcal{L}_{\text{cls}}(Z_s, G_s, Y_s) = \frac{1}{n_s} \sum_{i=1}^{n_s} \mathcal{L}_{\text{ce}}\left(F_{\text{lo}}\left(z_i^s\right), y_i^s\right) + \frac{1}{n_s} \sum_{i=1}^{n_s} \mathcal{L}_{\text{ce}}\left(F_{\text{to}}\left(g_i^s\right), y_i^s\right) \tag{21}$$

where $\mathcal{L}_{\text{ce}}$ represents the cross-entropy classification loss.

In general, although JAN-STF involves multiple subtraining steps, it differs from previous ADA methods that require multistage training [23], [28]. JAN-STF achieves end-to-end training by introducing balancing coefficients, significantly simplifying the entire training process while ensuring training stability, and thereby enhancing training efficiency. Therefore, the final optimization formulas for JAN-STF are summarized as follows:

$$\min_{\theta_D}(\alpha \mathcal{L}_{\text{discrim}}) \tag{22}$$

$$\min_{\theta_{F_{\text{lo}}}, \theta_{F_{\text{to}}}} \left(\mathcal{L}_{\text{cls}} - \gamma \mathcal{L}_{\text{discrep}}\right) \tag{23}$$

$$\min_{\theta_E}\left(\mathcal{L}_{\text{cls}} - \alpha \mathcal{L}_{\text{discrim}} + \beta \mathcal{L}_{\text{dist}} + \gamma \mathcal{L}_{\text{discrep}}\right) \tag{24}$$

where α, β, and γ are balance coefficients and specific configuration details will be discussed in Section IV-C. During the inference stage, the local classifier's classification results are directly used for final predictions. The pseudocode for the JAN-STF procedure is presented in Algorithm 1.

Two specific details require attention. First, the training parameters W and $\widetilde{W}$ of the topological feature extractor GraphSAGE are initialized randomly and updated in the same direction during training as the feature extractor by the same loss. Additionally, inspired by [21], both the intra-domain and the cross-domain graph network in JAN-STF are dynamically constructed in batches during the training process. In this manner, the intra-domain graph network can learn the topological structure information of the entire source or target domain through multiple rounds of batch random node sampling and information aggregation, thereby enhancing the domain topological representation of nodes in the cross-domain graph network. Simultaneously, the cross-domain graph network undergoes multiple rounds of random batch node sampling in the source and target domain, enabling effective communication of the overall topological structure information across domains.

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Algorithm 1 JAN-STF Training and Inference

Input: Source scene dataset X_s with corresponding labels Y_s, target scene dataset X_t, and training iterations N.

Output: The predicted labels Y_t for the target scene.

Initialize the parameters θ_E of the feature extractor E, $\theta_{F_{\text{lo}}}$ of the local classifier F_{lo}, $\theta_{F_{\text{to}}}$ of the topological classifier F_{to}, and θ_D of the domain discriminator D.
for $i \leftarrow 1$ to N **do**
  $Z_s, Z_t \leftarrow E(X_s), E(X_t)$
  Obtain semantic matrix p, pseudo-labels $\hat{Y}_t$ and cross-domain topological features $\widetilde{G}$ by Eqs. (2), (3), (4).
  Compute the discrimination loss $\mathcal{L}_{\text{discrim}}$ based on $\widetilde{G}$ using Eq. (7)
  Learn intra-domain topological features G_s, semantic matrix q and pseudo-labels $\check{Y}_t$ via Eqs. (9), (11), (12).
  Compute the distance loss $\mathcal{L}_{\text{dist}}$ via Eq. (14) based on Z_s, Z_t, Y_s, $\check{Y}_t$.
  $\bar{p}, \bar{q} \leftarrow softmax(p), softmax(q)$
  Compute the discrepancy loss $\mathcal{L}_{\text{discrep}}$ based on $\bar{p}$ and $\bar{q}$ using Eq. (18).
  Compute the classification loss $\mathcal{L}_{\text{cls}}$ based on Z_s, G_s and Y_s via Eq. (21).
  Update θ_D by $\mathcal{L}_{\text{discrim}}$ and Eq. (22).
  Update $\theta_{F_{\text{lo}}}$, $\theta_{F_{\text{to}}}$ by $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{discrep}}$ and Eq. (23).
  Update θ_E by $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{discrim}}$, $\mathcal{L}_{\text{dist}}$, $\mathcal{L}_{\text{discrep}}$ and Eq. (24).
end for
$Y_t \leftarrow F_{\text{lo}}(E(X_t))$.

IV. Experiments and Analysis

To validate the effectiveness of the proposed JAN-STF, we conduct experimental comparisons with nine other algorithms on three benchmark datasets of HSICC. Three evaluation metrics, namely, average accuracy (AA), overall accuracy (OA), and KC [44], are employed for performance assessment.

A. Dataset Description

1) Houston: The Houston dataset contains two scenes, Houston2013 [45] and Houston2018 [46]. The Houston2013 scene comprises 349 × 1905 pixels with 144 spectral bands ranging from 380 to 1050 nanometers. In contrast, the Houston2018 scene has 48 spectral bands within the same wavelength range. Both scenes cover seven land cover categories. For analysis, 48 spectral bands are selected from the Houston2013 scene to match the band settings of Houston2018, focusing on a 210 × 954 pixel overlapping region. Table I provides the sample counts for each category.

2) HyRANK: The HyRANK [47] dataset contains two scenes: Dioni and Loukia, with image dimensions of 250 × 1376 pixels and 249 × 945 pixels, respectively. These data consist of 176 spectral bands. Table II details category-specific sample counts.

3) Pavia: The Pavia dataset includes two scenes: Pavia University (PaviaU) and Pavia Center (PaviaC). PaviaU has an image size of 610 × 340 pixels with 103 spectral bands, and PaviaC has an image size of 1096 × 715 pixels with 102 spectral bands. To ensure consistency, the last band in PaviaU is removed, and seven common categories across PaviaU and PaviaC are selected for analysis. Table III presents the sample counts for each category.

TABLE I
Dataset Sample Quantity in Houston

No.	Class	Source (Houston2013)	Target (Houston2018)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total samples		2530	53200

TABLE II
Dataset Sample Quantity in HyRANK

No.	Class	Source (Dioni)	Target (Loukia)
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sderophyllous vegetation	5035	2996
8	Sparce sderophyllous vegetation	6374	2361
9	Sparcely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
Total samples		20024	10317

TABLE III
Dataset Sample Quantity in Pavia

No.	Class	Source (PaviaU)	Target (PaviaC)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total samples		39332	39355

B. Experimental Settings

All experiments in this study utilize PyTorch on platform with an RTX 2080Ti GPU and an i5-12490F CPU. Similar to the work [21], we utilize a fraction of source domain data (default 5%) along with an equal amount of unlabeled target domain data for training. Transfer classification results are then evaluated across the entire target domain figure. Each training sample undergoes data augmentation through random flipping and the introduction of noise, while no preprocessing is performed in testing. Input sample image blocks are sized at 12 × 12. Optimization involves batch stochastic gradient descent with a momentum of 0.9, $L2$-norm decay set to 1e-4, and a batch size of 100. A dynamic learning rate strategy is implemented with $\delta_\tau = \delta_0/((1+\varepsilon\tau)^\mu)$, where τ

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TABLE IV
STRUCTURES OF DISCRIMINATOR AND CLASSIFIERS

Layers	Discriminator	Local classifier	Topological classifier
1	$FC(out:512)$	$Dropout(0.5)$	$FC(out:1024)$
		$FC(out:4096)$	
	$LReLU(0.2)$	BN	$ReLU$
	$Dropout(0.5)$	$ReLU$	$Dropout(0.5)$
2	$FC(out:512)$	$Dropout(0.5)$	$FC(out:1024)$
		$FC(out:256)$	
	$LReLU(0.2)$	BN	$ReLU$
	$Dropout(0.5)$	$ReLU$	$Dropout(0.5)$
3	$FC(out:1)$	$Dropout(0.5)$	$FC(out:c)$
	$Sigmoid$	$FC(out:c)$	

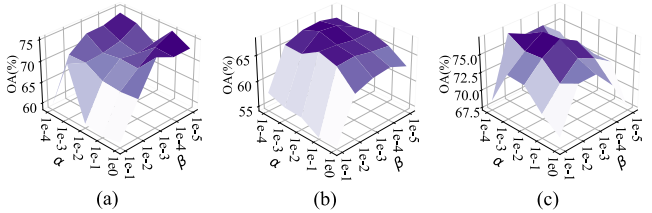


Fig. 7. OA performance of various combinations of α and β in JAN-STF across three datasets. (a) Houston. (b) HyRANK. (c) Pavia.

linearly increases from 0 to 1 during training, δ_0 is the base learning rate, ε is 10, and μ is 0.75. The feature extractor in JAN-STF employs the classic VGG16 network as detailed in [48]. JAN-STF comprises a domain discriminator and two classifiers. Specifically, there is a local classifier based on local features and a topological classifier based on topological features. The structural details are shown in Table IV.

In Table IV, FC stands for the fully connected layer, BN represents the batch normalization layer, LReLU denotes LeakyReLU, and c signifies the number of output classes. Table IV reveals that the dimensionality of features and the arrangement of activation functions in the local classifier and topological classifier vary according to the distinct characteristics of local and topological features. Each experiment comprises 300 iterations per round, repeated ten times for fair averaging and variance calculation, thereby minimizing biased estimates' impact on the final results. The study employs AA, OA, and KC as objective evaluation metrics for a comprehensive assessment.

C. Parameter Analysis and Settings

1) Balance Coefficients: JAN-STF achieves end-to-end training by balancing discriminative, distance, and discrepancy losses using coefficients α, β, and γ. Across all datasets, γ is consistently set to 1, while α and β are chosen from ranges $[1e-4{:}10{:}1e0]$ and $[1e-5{:}10{:}1e-1]$, respectively. Fig. 7 shows the mean OA over ten repetitions for various combinations of α and β, with default settings for other parameters.

Fig. 7 illustrates that JAN-STF exhibits suboptimal performance on the HyRANK and Pavia datasets when $\alpha < 1e0$, while achieving optimal performance only when $\alpha \geq 1e-1$ on the Houston dataset. This discrepancy arises from significant differences in sample quantities between Houston2013 and Houston2018, leading to less diverse topological relationships.

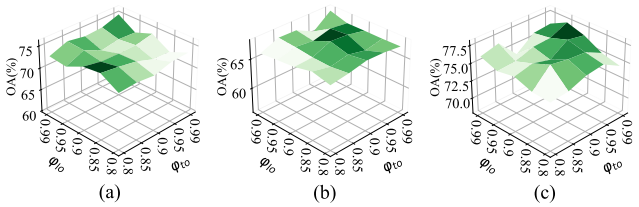


Fig. 8. OA performance of various combinations of φ_{lo} and φ_{to} in JAN-STF across three datasets. (a) Houston. (b) HyRANK. (c) Pavia.

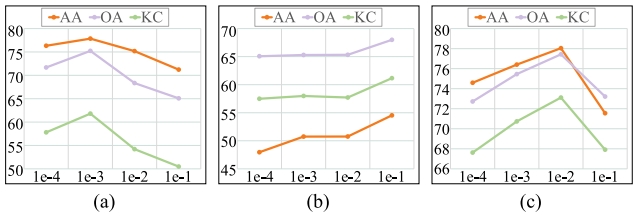


Fig. 9. OA performance of different δ_0 in JAN-STF across three distinct cross-scene datasets. (a) Houston. (b) HyRANK. (c) Pavia.

Additionally, JAN-STF demonstrates favorable β ranges on the Houston, HyRANK, and Pavia datasets as follows: $\beta < 1e-3$, $1e-4 < \beta < 1e-2$, and $1e-3 < \beta < 1e-2$. This is attributed to Houston's less complex sample categories, necessitating a reduction in the weight of category alignment loss to minimize the impact of mislabeled samples. Therefore, the default settings for the Houston, HyRANK, and Pavia datasets are ($\alpha = 1e-1$, $\beta = 1e-5$), ($\alpha = 1e-3$, $\beta = 1e-3$), and ($\alpha = 1e-2$, $\beta = 1e-3$), respectively.

2) Threshold Parameters: The threshold parameters φ_{lo} and φ_{to} play a crucial role in JAN-STF as they affect the semantic pseudo-labels, which are central to the algorithm's operation. Predicted scores undergo softmax preprocessing before being compared with a threshold to ensure maximum probability values per sample do not exceed 1. Therefore the threshold range is set as [0.8, 0.85, 0.90, 0.95, 0.99]. Fig. 8 illustrates the average OA across ten repetitions when evaluating different combinations of φ_{lo} and φ_{to} under default settings for other parameters.

From Fig. 8, it can be observed that JAN-STF performs optimally in Houston (indicated by deeper shades of green) with higher threshold combinations, while in HyRANK and Pavia, relatively lower threshold combinations are preferable. This is attributed to Houston's limited labeled data, necessitating strict thresholds for selecting high-quality pseudo-labels. Optimal threshold values for the Houston, HyRANK, and Pavia datasets are ($\varphi_{lo} = 0.99$, $\varphi_{to} = 0.99$), ($\varphi_{lo} = 0.90$, $\varphi_{to} = 0.90$), and ($\varphi_{lo} = 0.95$, $\varphi_{to} = 0.99$), respectively.

3) Base Learning Rate: To optimize the end-to-end training of JAN-STF, choosing the appropriate learning rate is essential. A range of $[1e-4{:}10{:}1e-1]$ for the base learning rate (δ_0) is considered. Then, testing these δ_0 values across datasets with default settings for other parameters, Fig. 9 shows the average OA over ten repetitions.

Fig. 9 shows that smaller learning rates enhance performance for the Houston dataset, while slightly larger rates expedite optimization for the larger HyRANK and Pavia datasets. Consequently, default learning rates of 1e−3, 1e−1,

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and 1e−2 are chosen for the Houston, HyRANK, and Pavia datasets, respectively.

D. Experimental Results and Analysis

The comparative algorithms comprise a source-only-based (So-based) method, CNN, five adversarial learning-based (Ad-based) methods: CDAN [36], MCD [28], JADA [34], BCDM [37], UJADA [35], one topology-based (To-based) method, TSTnet [21], and two hybrid-based (Hy-based) methods: BCAN [33] and LLURnet [15]. Specifically, BCAN integrates self-training with adversarial learning, while LLURnet amalgamates self-supervised with adversarial learning techniques. All of these methods have been introduced in the introduction and related work section. To ensure fair comparisons, all algorithms utilize the same extractor, namely VGG16 [48] which is widely employed in HSI. Regarding the parameter setting, the weight parameter λ increases nonlinearly from 0 to 1 during the training process in CDAN. The number of repeat updates n for MCD on the Houston, HyRANK, and Pavia datasets is 1, 5, and 1, respectively. In JADA, the weight parameter λ is set to 1, 5e−3, and 1e−2 on the three datasets. For BCDM, the weight parameter η is set to 1e−2. The weight parameters in TSTnet are ($\lambda_1 = 1e0$, $\lambda_2 = 1e1$), ($\lambda_1 = 1e0$, $\lambda_2 = 1e{-}1$), and ($\lambda_1 = 1e{-}1$, $\lambda_2 = 1e{-}1$) on the three datasets. In the UJADA method, the weight parameter β is set to 1e−2. In BCAN, the threshold parameter α is set to 0.90. The weight parameters λ and dimensions dim in the LLURnet method are set to ($\lambda_1 = 1e2$, $\lambda_2 = 1e{-}2$, $\text{dim}_1 = 16$, $\text{dim}_2 = 64$), ($\lambda_1 = 10$, $\lambda_2 = 100$, $\text{dim}_1 = 8$, $\text{dim}_2 = 16$), and ($\lambda_1 = 10$, $\lambda_2 = 100$, $\text{dim}_1 = 8$, $\text{dim}_2 = 16$) on the three datasets, respectively.

1) Performance on Houston Dataset: For cross-scene classification on Houston, Fig. 10 showcases JAN-STF predictions on Houston2018 alongside nine other algorithm results.

The results in Fig. 10 illustrate JAN-STF's superiority over other methods, exhibiting reduced noise and better alignment with actual land cover distribution. Particularly in the area marked by the white box (Fig. 10), where tree categories are often misclassified as grass by other algorithms, both TSTnet and JAN-STF accurately classify trees due to their incorporation of topological structure information. However, TSTnet still struggles with misclassifying non-residential buildings as roads, a challenge mitigated more effectively by JAN-STF. This advantage is attributed to JAN-STF's use of implicitly learned metrics through adversarial training, offering superior representational capabilities compared to TSTnet's explicitly defined distance metric function. Table V provides a quantitative comparison of the results.

It can be seen from Table V that domain adaptation methods significantly outperform source-only approaches like CNN, highlighting the efficacy of adversarial learning in deep domain adaptation. Furthermore, the joint adversarial algorithms JADA, composed of MCD and DANN [23], and UJADA, composed of BCDM and DANN, achieved OA performances of 72.65%–69.43%, surpassing those of MCD and BCDM. This emphasizes the synergy between domain-level and class-level adversarial strategies, enhancing transfer performance. The JAN-STF algorithm, leveraging the joint adversarial strategy, outperforms other algorithms in Houston2018 scene classification, improving AA, OA, and KC by 0.96%, 1.97%, and 3% respectively.

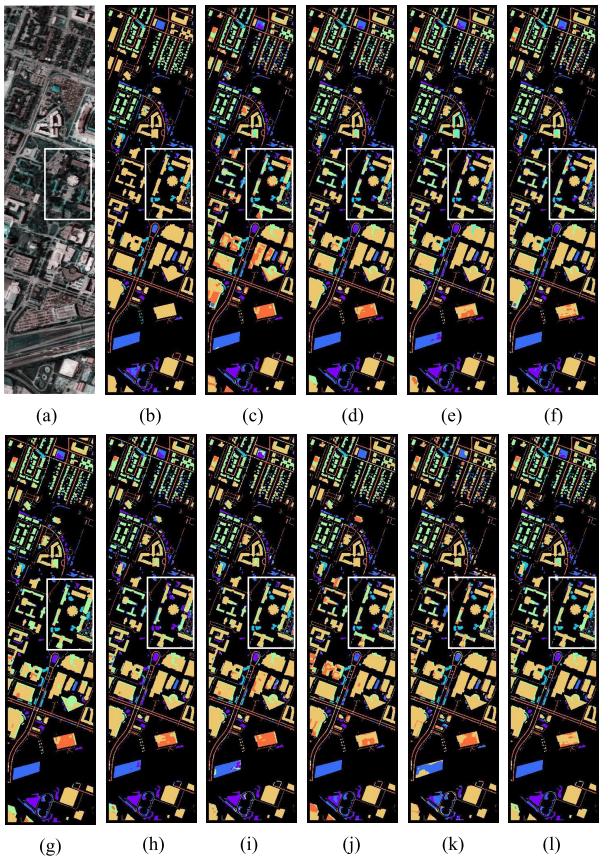


Fig. 10. Classification result maps for the target scene Houston2018. (a) False color image. (b) Ground truth. (c) CNN. (d) CDAN. (e) MCD. (f) JADA. (g) BCDM. (h) UJADA. (i) TSTnet. (j) BCAN. (k) LLURnet. (l) JAN-STF.

2) Performance on HyRANK Dataset: For the cross-scene classification task on the HyRANK dataset, JAN-STF's predictions on the target scene Loukia are shown in Fig. 11, along with results from nine other algorithms.

The classification results in Fig. 11 reveal the increased complexity of land cover categories in the HyRANK dataset, especially in the white-boxed region, making knowledge transfer more challenging. This region contains three highly similar categories: sparse sclerophyllous vegetation, sparsely vegetated areas, and rocky sand. Most algorithms struggle to differentiate these categories and often misclassify them as non-irrigated arable land. In contrast, JAN-STF utilizes topological structure information specific to these categories, achieving more accurate classification than other locally feature-based ADA methods. JAN-STF also exhibits lower noise levels compared to TSTnet, which incorrectly identifies sparse non-leafy vegetation as damaged grassland due to JAN-STF's more adaptable implicit metric for topology alignment. Table VI offers a quantitative comparison of the results.

Table VI reveals that for the HyRANK dataset, characterized by complex land cover distributions and similar spectral information among categories, joint adversarial strategies

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TABLE V

CLASSIFICATION ACCURACIES OF NINE METHODS ON TARGET DOMAIN (HOUSTON2018) USING 5% OF SOURCE DOMAIN (HOUSTON2013) SAMPLES

Class	So-based	Ad-based					To-based	Hy-based		JAN-STF
	CNN	CDAN	MCD	JADA	BCDM	UJADA	TSTnet	BCAN	LLURnet	
1	$63.92_{\pm 28}$	$81.77_{\pm 6.5}$	$86.65_{\pm 3.9}$	$79.34_{\pm 24}$	$93.11_{\pm 4.9}$	$90.03_{\pm 4.1}$	$84.01_{\pm 10}$	$69.56_{\pm 16}$	$32.15_{\pm 16}$	$83.75_{\pm 6.4}$
2	$79.43_{\pm 10}$	$74.41_{\pm 4.7}$	$74.07_{\pm 5.4}$	$71.74_{\pm 11}$	$60.12_{\pm 14}$	$68.90_{\pm 9.2}$	$67.56_{\pm 11}$	$82.82_{\pm 4.8}$	$73.58_{\pm 5.5}$	$75.10_{\pm 6.2}$
3	$56.51_{\pm 12}$	$68.03_{\pm 4.6}$	$60.53_{\pm 4.7}$	$66.29_{\pm 6.7}$	$62.72_{\pm 6.8}$	$63.80_{\pm 6.5}$	$53.12_{\pm 9.5}$	$57.40_{\pm 14}$	$62.83_{\pm 3.0}$	$67.74_{\pm 5.2}$
4	$85.61_{\pm 12}$	$95.45_{\pm 9.3}$	$98.64_{\pm 2.1}$	$95.45_{\pm 6.1}$	$94.55_{\pm 7.6}$	$99.09_{\pm 1.8}$	$47.73_{\pm 22}$	$63.18_{\pm 22}$	$65.00_{\pm 22}$	$98.18_{\pm 2.2}$
5	$62.46_{\pm 14}$	$76.13_{\pm 6.3}$	$79.01_{\pm 4.8}$	$75.72_{\pm 4.5}$	$74.53_{\pm 9.8}$	$81.95_{\pm 6.8}$	$69.77_{\pm 9.4}$	$80.49_{\pm 11}$	$73.54_{\pm 8.9}$	$78.38_{\pm 4.2}$
6	$53.47_{\pm 8.6}$	$72.94_{\pm 3.3}$	$71.15_{\pm 5.9}$	$74.71_{\pm 6.5}$	$65.98_{\pm 9.1}$	$67.54_{\pm 5.9}$	$74.14_{\pm 4.9}$	$76.07_{\pm 5.1}$	$78.72_{\pm 5.2}$	$77.04_{\pm 3.6}$
7	$64.30_{\pm 16}$	$67.22_{\pm 6.6}$	$63.67_{\pm 11}$	$61.50_{\pm 9.0}$	$70.08_{\pm 6.6}$	$66.93_{\pm 4.2}$	$57.89_{\pm 4.6}$	$53.14_{\pm 10}$	$39.82_{\pm 11}$	$64.73_{\pm 4.8}$
AA(%)	$66.53_{\pm 5.5}$	$76.57_{\pm 2.6}$	$76.25_{\pm 1.2}$	$74.97_{\pm 2.9}$	$74.44_{\pm 2.1}$	$76.89_{\pm 1.8}$	$64.89_{\pm 3.3}$	$60.81_{\pm 4.7}$	$68.95_{\pm 3.1}$	$\mathbf{77.85_{\pm 1.0}}$
OA(%)	$58.49_{\pm 3.3}$	$72.69_{\pm 2.6}$	$71.17_{\pm 1.2}$	$72.65_{\pm 2.9}$	$67.33_{\pm 4.7}$	$69.43_{\pm 3.9}$	$70.30_{\pm 3.3}$	$73.25_{\pm 3.0}$	$71.06_{\pm 1.9}$	$\mathbf{75.22_{\pm 1.6}}$
KC(%)	$43.21_{\pm 2.6}$	$58.80_{\pm 1.9}$	$56.96_{\pm 3.1}$	$58.36_{\pm 3.0}$	$52.38_{\pm 4.9}$	$55.39_{\pm 4.4}$	$54.51_{\pm 3.9}$	$58.16_{\pm 3.9}$	$52.69_{\pm 2.3}$	$\mathbf{61.80_{\pm 1.8}}$

TABLE VI

CLASSIFICATION ACCURACIES OF NINE METHODS ON TARGET DOMAIN (LOUKIA) USING 5% OF SOURCE DOMAIN (DIONI) SAMPLES

Class	So-based	Ad-based					To-based	Hy-based		JAN-STF
	CNN	CDAN	MCD	JADA	BCDM	UJADA	TSTnet	BCAN	LLURnet	
1	$32.67_{\pm 16}$	$21.31_{\pm 11}$	$35.73_{\pm 7.4}$	$25.78_{\pm 9.5}$	$40.87_{\pm 9.5}$	$17.14_{\pm 11}$	$31.21_{\pm 12}$	$12.14_{\pm 11}$	$02.37_{\pm 2.4}$	$29.76_{\pm 6.9}$
2	$47.22_{\pm 34}$	$63.15_{\pm 21}$	$37.22_{\pm 32}$	$53.15_{\pm 30}$	$41.85_{\pm 32}$	$30.37_{\pm 35}$	$28.33_{\pm 27}$	$31.85_{\pm 32}$	$18.72_{\pm 33}$	$56.85_{\pm 31}$
3	$38.08_{\pm 23}$	$28.17_{\pm 15}$	$28.08_{\pm 19}$	$30.42_{\pm 15}$	$36.36_{\pm 13}$	$07.65_{\pm 7.2}$	$30.63_{\pm 9.7}$	$08.54_{\pm 11}$	$09.73_{\pm 15}$	$40.31_{\pm 19}$
4	$23.16_{\pm 22}$	$33.29_{\pm 22}$	$08.35_{\pm 9.3}$	$24.56_{\pm 19}$	$21.65_{\pm 9.9}$	$01.01_{\pm 2.3}$	$11.90_{\pm 18}$	$07.09_{\pm 14}$	$09.00_{\pm 8.3}$	$20.76_{\pm 18}$
5	$53.37_{\pm 20}$	$58.90_{\pm 17}$	$71.45_{\pm 19}$	$64.58_{\pm 11}$	$59.24_{\pm 13}$	$56.87_{\pm 12}$	$65.32_{\pm 13}$	$17.52_{\pm 14}$	$24.42_{\pm 25}$	$73.62_{\pm 10}$
6	$28.82_{\pm 13}$	$22.30_{\pm 15}$	$17.89_{\pm 12}$	$11.30_{\pm 14}$	$25.59_{\pm 12}$	$11.42_{\pm 6.1}$	$04.08_{\pm 3.3}$	$20.62_{\pm 13}$	$22.85_{\pm 16}$	$19.31_{\pm 18}$
7	$66.08_{\pm 2.4}$	$70.44_{\pm 4.8}$	$69.79_{\pm 5.9}$	$69.11_{\pm 2.7}$	$67.16_{\pm 4.3}$	$67.76_{\pm 4.2}$	$73.17_{\pm 4.9}$	$85.34_{\pm 5.2}$	$80.47_{\pm 9.7}$	$68.26_{\pm 2.7}$
8	$70.69_{\pm 8.5}$	$71.76_{\pm 6.4}$	$64.20_{\pm 5.9}$	$74.21_{\pm 7.4}$	$66.38_{\pm 5.1}$	$66.75_{\pm 2.8}$	$66.93_{\pm 5.8}$	$71.01_{\pm 5.6}$	$64.81_{\pm 5.8}$	$70.84_{\pm 6.5}$
9	$50.23_{\pm 20}$	$57.32_{\pm 16}$	$79.07_{\pm 13}$	$65.79_{\pm 14}$	$75.54_{\pm 12}$	$77.09_{\pm 12}$	$78.97_{\pm 7.3}$	$57.84_{\pm 19}$	$71.43_{\pm 18}$	$55.51_{\pm 21}$
10	$03.38_{\pm 6.4}$	$07.40_{\pm 22}$	$01.66_{\pm 3.7}$	$00.22_{\pm 0.6}$	$00.13_{\pm 0.4}$	$30.35_{\pm 34}$	$05.17_{\pm 12}$	$00.00_{\pm 0.0}$	$30.37_{\pm 30}$	$19.38_{\pm 28}$
11	$100.0_{\pm 0.0}$	$99.96_{\pm 0.1}$	$100.0_{\pm 0.0}$	$99.94_{\pm 0.0}$	$100.0_{\pm 0.0}$	$100.0_{\pm 0.0}$	$99.93_{\pm 0.2}$	$100.0_{\pm 0.0}$	$100.0_{\pm 0.0}$	$99.83_{\pm 0.5}$
12	$100.0_{\pm 0.0}$	$100.0_{\pm 0.0}$	$100.0_{\pm 0.0}$	$100.0_{\pm 0.2}$	$99.74_{\pm 0.8}$	$100.0_{\pm 0.0}$	$100.0_{\pm 0.0}$	$97.96_{\pm 3.6}$	$99.52_{\pm 1.3}$	$99.95_{\pm 0.1}$
AA(%)	$51.14_{\pm 5.6}$	$52.38_{\pm 2.6}$	$51.12_{\pm 4.6}$	$51.95_{\pm 2.7}$	$52.88_{\pm 4.5}$	$47.20_{\pm 3.1}$	$49.64_{\pm 2.9}$	$42.49_{\pm 2.9}$	$44.51_{\pm 4.1}$	$\mathbf{54.53_{\pm 4.8}}$
OA(%)	$64.60_{\pm 1.6}$	$66.40_{\pm 1.7}$	$66.20_{\pm 1.9}$	$66.81_{\pm 1.4}$	$65.30_{\pm 1.3}$	$64.24_{\pm 1.2}$	$66.72_{\pm 1.4}$	$64.31_{\pm 1.8}$	$64.04_{\pm 1.5}$	$\mathbf{67.99_{\pm 1.8}}$
KC(%)	$56.97_{\pm 2.1}$	$59.02_{\pm 2.3}$	$59.07_{\pm 2.6}$	$59.48_{\pm 1.6}$	$58.03_{\pm 1.7}$	$56.78_{\pm 1.6}$	$59.66_{\pm 1.8}$	$54.99_{\pm 2.4}$	$55.30_{\pm 1.9}$	$\mathbf{61.15_{\pm 2.0}}$

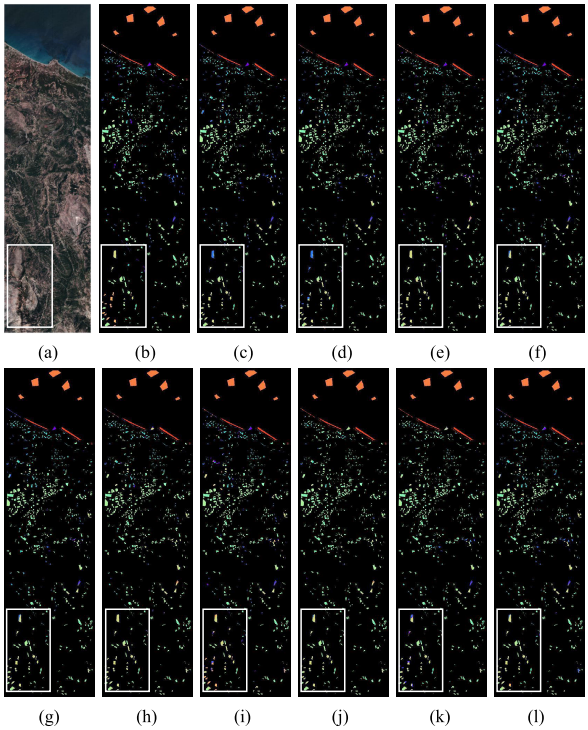


Fig. 11. Classification result maps for the target scene Loukia. (a) False color image. (b) Ground truth. (c) CNN. (d) CDAN. (e) MCD. (f) JADA. (g) BCDM. (h) UJADA. (i) TSTnet. (j) BCAN. (k) LLURnet. (l) JAN-STF.

may not consistently outperform individual class-level methods. For instance, UJADA shows an OA 1.06% lower than BCDM, possibly due to CNN's limitations in capturing nuanced local features, leading to potential alignment challenges with more complex adversarial networks. This underscores the importance of comprehensive feature representation for robust domain adaptation. TSTnet constructs a topological distance metric without adversarial learning, yet it surpasses many ADA methods relying solely on local features, with an OA value only 0.09% lower than the suboptimal adversarial method, JADA. This highlights the compensatory role of topological relationships in addressing local feature alignment limitations. Unlike TSTnet, which relies solely on intra-domain topological features for domain adaptation, JAN-STF incorporates cross-domain topological features into the adversarial learning process, significantly improving alignment performance across domains. As a result, JAN-STF outperforms other top-performing algorithms by 1.65%, 1.80%, and 1.49% in terms of AA, OA, and KC, respectively.

3) Performance on Pavia Dataset: For the cross-scene classification task on the Pavia dataset, JAN-STF's predictions on the target scene PaviaC are shown in Fig. 12, along with results from nine other algorithms.

The predicted result images in Fig. 12 exhibit noticeable noise, indicating classification inaccuracies likely due to significant spectral variations for the same categories across scenes in the Pavia dataset. However, JAN-STF maintains superior performance despite these challenges. For instance, within the white box region in Fig. 12, many algorithms using

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TABLE VII

CLASSIFICATION ACCURACIES OF NINE METHODS ON TARGET DOMAIN (PAVIAC) USING 5% OF SOURCE DOMAIN (PAVIAU) SAMPLES

Class	So-based	Ad-based					To-based	Hy-based		JAN-STF
	CNN	CDAN	MCD	JADA	BCDM	UJADA	TSTnet	BCAN	LLURnet	
1	$65.61_{\pm 5.0}$	$62.68_{\pm 6.6}$	$52.37_{\pm 14}$	$65.91_{\pm 13}$	$49.15_{\pm 16}$	$60.66_{\pm 21}$	$67.67_{\pm 9.9}$	$69.28_{\pm 19}$	$77.30_{\pm 21}$	$76.02_{\pm 8.6}$
2	$84.26_{\pm 7.7}$	$90.89_{\pm 3.5}$	$89.30_{\pm 7.7}$	$89.59_{\pm 4.4}$	$87.16_{\pm 7.6}$	$85.28_{\pm 6.0}$	$89.39_{\pm 3.5}$	$81.28_{\pm 6.8}$	$93.04_{\pm 1.9}$	$87.86_{\pm 5.8}$
3	$58.57_{\pm 10}$	$62.31_{\pm 3.5}$	$43.89_{\pm 7.6}$	$44.96_{\pm 4.4}$	$42.43_{\pm 19}$	$58.18_{\pm 22}$	$63.39_{\pm 9.3}$	$35.02_{\pm 28}$	$70.38_{\pm 9.3}$	$65.90_{\pm 15}$
4	$50.29_{\pm 15}$	$54.04_{\pm 12}$	$43.29_{\pm 26}$	$38.39_{\pm 22}$	$49.87_{\pm 16}$	$59.95_{\pm 12}$	$47.81_{\pm 9.7}$	$49.83_{\pm 24}$	$71.41_{\pm 12}$	$57.31_{\pm 6.9}$
5	$87.58_{\pm 2.6}$	$86.45_{\pm 4.9}$	$74.03_{\pm 6.4}$	$73.84_{\pm 10}$	$72.33_{\pm 7.0}$	$72.80_{\pm 6.8}$	$91.00_{\pm 6.6}$	$83.89_{\pm 12}$	$76.57_{\pm 4.6}$	$92.44_{\pm 3.3}$
6	$90.68_{\pm 2.6}$	$89.49_{\pm 2.5}$	$94.18_{\pm 2.9}$	$89.90_{\pm 5.9}$	$95.78_{\pm 2.8}$	$95.87_{\pm 2.3}$	$90.70_{\pm 2.8}$	$89.07_{\pm 8.9}$	$75.66_{\pm 7.9}$	$85.91_{\pm 1.7}$
7	$68.88_{\pm 9.9}$	$75.47_{\pm 12}$	$70.14_{\pm 13}$	$66.73_{\pm 23}$	$66.96_{\pm 17}$	$57.40_{\pm 21}$	$60.04_{\pm 13}$	$61.55_{\pm 29}$	$46.72_{\pm 13}$	$80.86_{\pm 13}$
AA(%)	$72.27_{\pm 3.1}$	$74.47_{\pm 3.8}$	$66.74_{\pm 2.8}$	$67.05_{\pm 3.5}$	$66.24_{\pm 4.3}$	$70.02_{\pm 4.8}$	$72.86_{\pm 4.4}$	$67.13_{\pm 4.2}$	$73.01_{\pm 3.8}$	$\mathbf{78.04_{\pm 1.6}}$
OA(%)	$70.79_{\pm 2.9}$	$73.66_{\pm 3.7}$	$66.62_{\pm 3.2}$	$67.55_{\pm 3.9}$	$66.08_{\pm 3.8}$	$69.25_{\pm 5.1}$	$71.03_{\pm 4.3}$	$67.48_{\pm 5.4}$	$74.14_{\pm 4.8}$	$\mathbf{77.43_{\pm 2.9}}$
KC(%)	$65.33_{\pm 3.5}$	$68.60_{\pm 4.7}$	$60.40_{\pm 3.8}$	$61.33_{\pm 4.6}$	$59.94_{\pm 4.3}$	$63.64_{\pm 5.7}$	$65.59_{\pm 5.0}$	$61.32_{\pm 6.3}$	$69.04_{\pm 5.6}$	$\mathbf{73.11_{\pm 3.2}}$

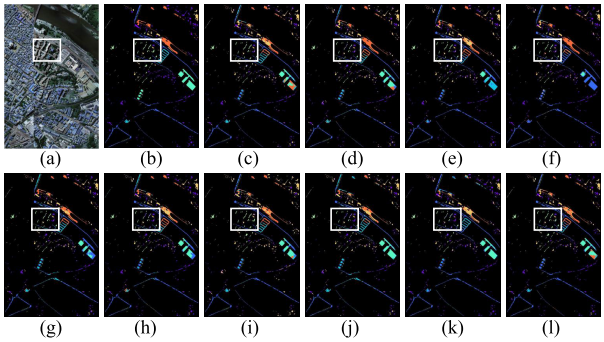


Fig. 12. Classification result maps for the target scene PaviaC. (a) Ground truth. (b) CNN. (c) CDAN. (d) MCD. (e) JADA. (f) BCDM. (g) UJADA. (h) TSTnet. (i) BCAN. (j) LLURnet. (k) JAN-STF.

ADA misclassify shadows as trees, with some even performing worse than CNN without adaptation. In contrast, JAN-STF reduces negative alignment risks by integrating semantic knowledge supervision. This enables JAN-STF to accurately classify shadows and correctly identify asphalt, which is often misjudged by other algorithms. Detailed experimental results are provided in Table VII.

Due to the significant spectral distribution differences between the PaviaU and PaviaC scenes, adversarial learning can easily break down, leading to negative alignment. Observing Table VII reveals that some ADA methods perform even worse than methods relying solely on source domain data. For instance, UJADA has an OA value 1.54% lower than the CNN method using only the source domain, possibly attributed to the absence of semantic guidance in the adversarial learning process leading to negative feature alignment. In contrast, LLURnet addresses this limitation by employing inter-class and intra-class contrastive learning, achieving a suboptimal OA of 74.14%. JAN-STF employs a topological semantic-aware mechanism to generate semantic pseudo-labels, narrowing the distance between cross-domain same-category features. Unlike BCAN, which treats semantic pseudo-labels as hard labels, JAN-STF computes distance loss with pseudo-labels, preserving the supervisory role of semantic labels while mitigating the impact of incorrect labels on model efficacy. Experimental results demonstrate JAN-STF outperforming other algorithms by 3.57%, 3.29%, and 4.07% in AA, OA, and KC metrics, respectively, even in challenging Pavia scenarios.

4) Performance Across Diverse Sample Ratios: This section conducts an evaluation of JAN-STF's robustness across a range of source domain sample percentages by juxtaposing its performance against four other prominent HSICC methods across three cross-scene datasets. The initial assessment focuses on sample sizes ranging from 10% to 100% of the Houston dataset, characterized by a comparatively limited source domain data volume. The average OA results of ten repeated experiments of this analysis are systematically detailed and presented in Table VIII.

TABLE VIII

OA RESULTS OF FIVE METHODS ACROSS VARYING RATIOS OF TRAINING SAMPLES FOR THE HOUSTON DATASETS

Methods	Sample ratio				
	10%	20%	40%	80%	100%
TSTnet	74.37	79.00	78.68	78.33	79.48
UJADA	75.63	76.20	77.51	77.61	78.17
BCAN	77.29	79.15	78.79	80.15	79.28
LLURnet	73.29	77.11	78.00	78.55	79.00
JAN-STF	**78.22**	**79.25**	**79.50**	**80.23**	**80.45**

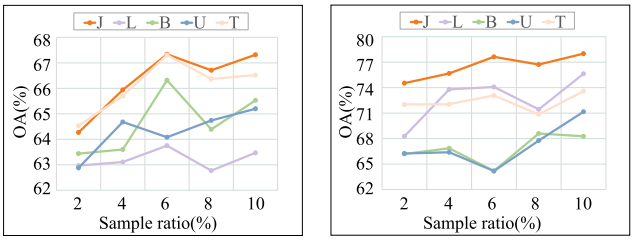


Fig. 13. OA results of five methods across varying ratios of training samples for the two datasets. J, L, B, U, and T, respectively, represent the JAN-STF, LLURnet, BCAN, UJADA, and TSTnet methods. (a) HyRANK. (b) Pavia.

From Table VIII, it is evident that as the percentage increases, the performance of JAN-STF steadily improves, consistently surpassing the other four algorithms at all sample ratios. For instance, at 100% sample ratio, BCAN experiences a 0.87% OA decrease due to increased mislabeled target instances compared to 80% sample ratio. In contrast, JAN-STF mitigates this impact through techniques like topological awareness, resulting in a 0.22% performance increase and surpassing the optimal method by 0.97%. Furthermore, to enhance the comprehensiveness of our experiments, we conduct experiments on the HyRANK and Pavia datasets with larger source domain data sizes, using sampling rates ranging from 2% to 10%. The results are depicted in Fig. 13.

Fig. 13 demonstrates a general upward OA trend as the number of source domain samples increases across five

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TABLE IX
IMPACT OF THREE CONTRIBUTIONS ON OA (%). C, T, AND L REPRESENT THE CGCAN, TADCA, AND LTCAN

Exp	C	T	L	Houston	HyRANK	Pavia
E1	×	×	×	$58.39_{\pm 4.9}$	$65.08_{\pm 1.2}$	$68.17_{\pm 3.4}$
E2	✓	×	×	$72.54_{\pm 2.0}$	$65.59_{\pm 2.0}$	$73.58_{\pm 3.5}$
E3	✓	✓	×	$74.10_{\pm 1.7}$	$67.29_{\pm 1.0}$	$75.27_{\pm 2.9}$
E4	✓	×	✓	$72.85_{\pm 1.7}$	$66.13_{\pm 1.4}$	$72.38_{\pm 2.9}$
E5	✓	✓	✓	$\mathbf{75.22_{\pm 1.6}}$	$\mathbf{67.99_{\pm 1.8}}$	$\mathbf{77.43_{\pm 2.9}}$

TABLE X
IMPACT OF SEMANTIC AND TOPOLOGY ON OA (%). SEM AND TOPO REPRESENT SEMANTIC AND TOPOLOGY

Exp	Sem	Topo	Houston	HyRANK	Pavia
A1	×	×	$71.89_{\pm 3.1}$	$65.40_{\pm 1.5}$	$65.67_{\pm 4.4}$
A2	✓	×	$73.38_{\pm 1.5}$	$66.45_{\pm 1.2}$	$72.75_{\pm 2.9}$
A3	×	✓	$73.17_{\pm 2.8}$	$66.28_{\pm 1.5}$	$74.73_{\pm 3.0}$
A4	✓	✓	$\mathbf{75.22_{\pm 1.6}}$	$\mathbf{67.99_{\pm 1.8}}$	$\mathbf{77.43_{\pm 2.9}}$

methods. On the Pavia dataset, JAN-STF outperforms the other four algorithms significantly. Nonetheless, on the HyRANK dataset, which has more classes and closer inter-class similarity, JAN-STF performs comparably to the topology-based TSTnet method before the 6% sampling ratio. However, the increase in data quantity may lead to the increased complexity of the samples and JAN-STF exhibits stronger robustness after the 6% sampling rate. At 8% sampling rate, TSTnet experiences a 0.94% OA decrease in performance, whereas JAN-STF only shows a 0.63% decrease. Furthermore, with a sample size increase to 10%, JAN-STF's OA value rebounds and surpasses TSTnet by a noticeable 0.8%. This enhancement is attributed to JAN-STF's construction of a more precise cross-domain topological structure for information aggregation from source and target domains. Thus, to rigorously assess the JAN-STF's performance under a small labeled sample size, and refer to the work [21], we opt for a default sampling rate of 5%.

E. Ablation Study

1) Analysis of the Role of Three Contributions: The three contributions of this article are as follows: proposing domain-level adversarial learning based on semantic-guided cross-domain topological information (CGCAN), introducing a dynamic category alignment mechanism based on topology-aware (TADCA), and presenting a class-level adversarial mechanism based on topological-local classifiers (LTCAN). To verify the role of these three contributions in improving the effectiveness of JAN-STF, we conduct five ablation experiments, namely $E1$, $E2$, $E3$, $E4$, and $E5$. $E1$ stands for no utilization of the three contributions, $E2$ for employing CGCAN, $E3$ for adopting both CGCAN and TADCA, $E4$ for integrating CGCAN and LTCAN, and $E5$ for concurrently integrating all three contributions. The results are presented in Table IX.

Table IX shows that $E2$ achieves higher OA results compared to $E1$ on the Houston, HyRANK, and Pavia datasets by 14.15%, 0.53%, and 5.41%, respectively. This demonstrates the effectiveness of cross-domain graph convolutional adversarial training in learning domain-invariant representations to reduce domain bias. Subsequently, $E3$ outperforms $E2$ on the three datasets by 1.56%, 1.7%, and 1.69%, highlighting the positive impact of reducing category distances through topological awareness. Furthermore, $E5$ surpasses $E3$ by 1.12%, 0.70%, and 2.16% on the three datasets, respectively. This indicates that utilizing class-level adversarial learning based on the discrepancy in topological-local classifiers can capture more target domain information further away from the source support boundaries, thereby enhancing classification performance in the target scene. Moreover, $E5$ exceeds $E4$ with OA increases of 2.37%, 1.86%, and 5.05% across the three datasets, highlighting the clear effectiveness of TADCA in maintaining the stability of class-level adversarial learning. In summary, these three contributions have played a crucial role in enhancing the domain adaptation performance of JAN-STF.

2) Analysis of the Role of Semantic and Topology: The most critical factors in JAN-STF are semantic and topology information. To assess the impact of semantic and topology on JAN-STF's performance, we conduct four ablation experiments, namely $A1$, $A2$, $A3$, and $A4$. $A1$ represents the removal of all semantic and topological information from JAN-STF, meaning the elimination of all semantic pseudo-labels and the substitution of topological features with local features. $A2$ is based solely on semantic, $A3$ solely on topology, and $A4$ combines both semantic and topological information in JAN-STF. The results are shown in Table X.

In Table X, $A2$ outperforms $A1$ in OA on the three datasets by 1.49%, 1.05%, and 7.08%, respectively. This highlights that semantic pseudo-labels can effectively enhance the discrimination of domain-invariant features. Moreover, $A3$ surpasses $A1$ in OA on the three datasets by 1.28%, 0.88%, and 9.06%, respectively. This affirms that integrating topological structure effectively overcomes the constraint of relying only on local information for knowledge transfer. $A4$ achieves higher OA performance compared to optimal performance in $A2$ and $A3$ by 1.84%, 1.54%, and 2.7% on the three datasets, respectively. This enhancement is attributed to semantic pseudo-labels aiding in constructing more accurate topological relationships, while topological features generate more reliable semantic pseudo-labels, leading to a more precise domain adaptation.

F. Convergence Performance

The convergence of the two min-max alternating optimization processes delineated in (22)–(24), concerning the feature extractor and discriminator, as well as the feature extractor and classifiers, has been empirically substantiated in prior studies such as [23] and [28]. Building upon this foundation, our approach draws inspiration from the techniques proposed in [34], incorporating gradient reversal layers and loss balancing coefficients into our framework. These enhancements are strategically designed to guarantee the convergence of the joint optimization process involving the aforementioned two min-max adversarial components. Specifically, gradient reversal layers can better balance the competition in feature

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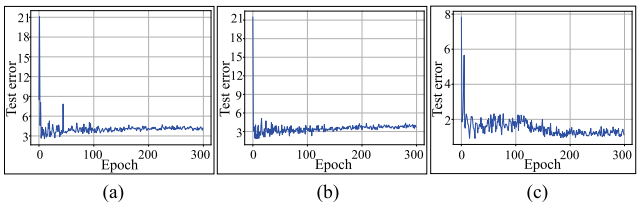


Fig. 14. Convergence performance for three cross-scene datasets. (a) Houston. (b) HyRANK. (c) Pavia.

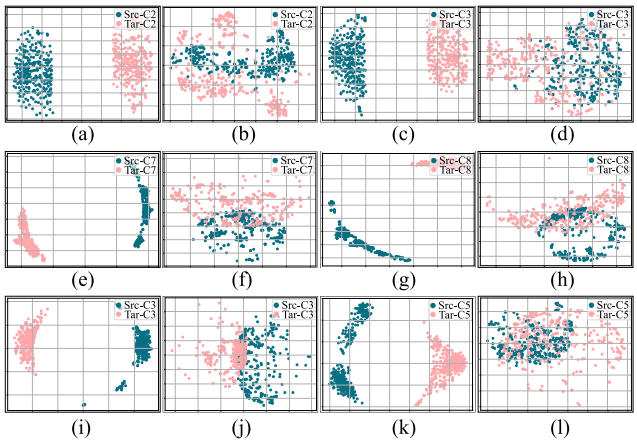


Fig. 15. Alignment performance of the proposed JAN-STF using the Houston, HyRANK, and Pavia datasets (IS-initial samples and AS-aligned samples). Src represents the source domain, and Tar represents the target domain. (a) IS from Houston (Class 2). (b) AS by JAN-STF from Houston (Class 2). (c) IS from Houston (Class 3). (d) AS by JAN-STF from Houston (Class 3). (e) IS from HyRANK (Class 7). (f) AS by JAN-STF from HyRANK (Class 7). (g) IS from HyRANK (Class 8). (h) AS by JAN-STF from HyRANK (Class 8). (i) IS from Pavia (Class 3). (j) AS by JAN-STF from Pavia (Class 3). (k) IS from Pavia (Class 5). (l) AS by JAN-STF from Pavia (Class 5).

extractor, discriminator and classifiers, enhancing the model's adversarial robustness, and leading to improved convergence in adversarial learning. Furthermore, balancing coefficients are utilized to adjust multiple losses, guiding the network toward stable optimization and preventing divergence. To verify the convergence performance of the proposed JAN-STF, we provide the test error curves on three cross-scene datasets, as shown in Fig. 14.

From Fig. 14, it can be seen that JAN-STF converges relatively quickly on all three datasets and reaches a stable state after 100 epochs.

G. Alignment Performance

To further demonstrate the excellent cross-domain alignment capability of JAN-STF, we utilize the t-SNE [49] algorithm to visualize the features of certain class samples extracted by the feature extractor from the Houston, HyRANK, and Pavia datasets. The visualization results are depicted in Fig. 15.

Fig. 15(a), (c), (e), (g), (i), and (k) clearly demonstrates the notable domain shift phenomenon in the initial samples of the Houston, HyRANK, and Pavia datasets, where cross-domain features exhibit significant differences even within the same class distribution. The proposed JAN-STF incorporates semantic information and topological structure information into the adversarial mechanism, enabling the feature extractor to learn to map the source domain and target domain into nearby spaces, thereby achieving alignment of cross-domain features within the same class. The results in Fig. 15(b), (d), (f), (h), (j), and (l) illustrate that the feature distributions of the source domain and target domain show significant overlaps after alignment by JAN-STF. This highlights the excellent domain discrepancy reduction capabilities of JAN-STF. Taking the third category of the Houston dataset as an example, Fig. 15(c) presents the initial feature distributions of the source domain and target domain, which exhibit no overlap and are considerably distant from each other. In Fig. 15(d), the feature distributions of the source domain and target domain after alignment are shown, revealing an almost complete overlap between the two.

V. Conclusion

JAN-STF is proposed to address poor transfer performance in HSICC, resulting from a narrow focus on local features in existing domain adaptation methods. First, domain-level adversarial learning is employed through CGCAN to acquire highly transferable domain-invariant representations. Subsequently, category alignment is achieved via TADCA to map cross-domain category features to nearby space, ensuring stability during the following class adversarial procedure. Finally, LTCAN enhances the discrimination performance of domain-invariant representation to address deficiencies from domain-level adversarial learning. Observations from classification result images indicate that JAN-STF more accurately distinguishes similar features of land cover categories, with predicted results closer to the actual category distribution in real-world scenes. Overall, JAN-STF achieves OA metrics of 75.22%, 67.99%, and 77.43% on the Houston, HyRANK, and Pavia datasets, respectively, surpassing the best-performing comparative algorithms by 1.97%, 1.18%, and 3.77%.

However, the optimal threshold values vary from dataset to dataset in this work, making hyperparameter tuning difficult. In the future, we will focus on finding a dynamic thresholding approach to eliminate the hassle of setting a fixed threshold for each dataset.

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